

Done last week:

1, 2, 3, 4, 5, 6, 7, 8, 9, 16.

Q10 Lecture 6

Q11 Lecture 6

Q14 Lecture 10

Q15 Lecture 10

Q17 Lecture 12

Q24 Lecture 17

Q25 Lecture 17

Q26 Lecture 17

Q26 Lecture 18

Q28 Lecture 18

Q30 Lecture 18

Q31 Lecture 19

Q32 Not on course

To do:

~~12~~, ~~13~~, 18, ~~19~~, 20, 21, 22, 23, 29, 33

Q12 A L -bit LFSR is a mechanism that produces an infinite sequence of binary digits from an initial binary list

$$s_0, s_1, \dots, s_{L-1}$$

it can be described by its connection polynomial

$$C(x) = 1 + c_1x + c_2x^2 + \dots + c_Lx^L$$

with $c_i \in \mathbb{Z}_2$ and the formula

$$s_j = c_1 s_{j-1} + c_2 s_{j-2} + \dots + c_L s_{j-L}$$

$$L = 4$$

$$C(x) = 1 + x + x^3$$

s_0	s_1	s_2	s_3	s_4	s_5	s_6	s_7	s_8	s_9	s_{10}
0	0	1	1	1	0	1	0	0	1	1

$$s_j = s_{j-1} + s_{j-3}$$

period 7

Q13

$$\begin{array}{r} p = 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 1 \ 1 \mid 0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 1 \\ + \\ s = 1 \ 0 \ 1 \ 0 \ 0 \ 1 \ 1 \ 1 \mid 0 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0 \\ || \\ c = 1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \mid 0 \ 0 \ 0 \ 1 \ 1 \ 0 \ 0 \end{array}$$

$$L=3$$

$$C(x) = 1 + c_1 x + c_2 x^2 + c_3 x^3$$

$$\begin{array}{cccccc} s_0 & s_1 & s_2 & s_3 & s_4 & s_5 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}$$

$$s_3 = 0 = c_1 s_2 + c_2 s_1 + c_3 s_0$$

$$\boxed{0 = c_1 + c_3}$$

$$s_4 = 0 = c_1 s_3 + c_2 s_2 + c_3 s_1$$

$$\boxed{0 = c_2}$$

$$s_5 = 1 = c_1 s_4 + c_2 s_3 + c_3 s_2$$

$$\boxed{1 = c_3}$$

$$C(x) = 1 + x + x^3$$

Q19 Customer has a "calculator" with a key

$$K = (N, e)$$

installed and a 2-digit Counter $c = c_1 c_2$ which increases by +1 each time the calculator is used.

you enter your PIN

$P_1 P_2 P_3 P_4$

and the calculator computes

$$m = (P_1 P_2 P_3 P_4 c_1 c_2)^e \bmod N$$

you type m into the online web page.

The bank knows your pin

$P_1 P_2 P_3 P_4$ and $K = (N, e)$, and

your last value $c-1$ of the Counter.

Hence bank checks that

$$(p_1 p_2 p_3 p_4 c_1 c_2)^e \equiv m \pmod{N}$$

for $c_1 c_2 = c, c+1, c+2, c+3$.